

General Characteristics

81. The number of unpaired electrons in cobalt atom is (atomic number of Co = 27)
 (a) 2 (b) 3
 (c) 4 (d) 1
82. Zn is related to which group
 (a) IIB (b) IIA
 (c) IA (d) IB
83. Which of the following element does not show variable valency
 (a) Ni (b) Zn
 (c) Cu (d) Mn
84. Which of the following is diamagnetic transitional metal ion
 (a) Ni^{+2} (b) Zn^{+2}
 (c) Co^{+2} (d) Cu^{+2}
85. Which of the following is not an actinide
 (a) Curium
 (b) Californium
 (c) Uranium
 (d) Terbium
86. The ability to form complex compounds by the transitional metal ion is due to
 (a) Small size
 (b) Vacant 'd' orbitals
 (c) High nuclear charge
 (d) All of these
87. Which of the following will give green hydrated ion
 (a) Fe^{+2} (b) Ni^{+2}
 (c) (a) and (b) both (d) V^{+3}
88. Magnetic moment is expressed in
 (a) Faraday (b) Calorie
 (c) Bohr Magneton (d) Debye
89. If the colours of salts of transition elements are due to the presence of unpaired electrons in the transition metal ions, which of the following ions will be colourless in aqueous solution
 (a) Ti^{3+} (b) Ti^{4+}
 (c) Fe^{2+} (d) Fe^{3+}
 (The at. no. of Ti and Fe are 22 and 26 respectively)
90. In the periodic table first transition series is placed in
 (a) Third period
 (b) Fourth period
 (c) Fifth period
 (d) None of these
91. The element having general electronic configuration $3d^4 4s^1$ is
 (a) Noble gas
 (b) Non-metal
 (c) Metalloid
 (d) Transition metal
92. Which of the following general configuration of outermost shell represents chromium element [Cr's atomic number = 24]
 (a) $d^5 s^1$ (b) $d^6 s^0$
 (c) $d^4 s^2$ (d) $d^3 s^2$



93. Which element gives maximum balanced oxide
(a) V (b) Cr
(c) Mn (d) Fe
94. In first transition series, the melting point of Mn is low because
(a) Due to d^{10} configuration, metallic bonds are strong
(b) Due to d^7 configuration, metallic bonds are weak
(c) Due to d^5 configuration, metallic bonds are weak
(d) None of these
95. Which of the following ions has the least magnetic moment
(a) Cu^{+2} (b) Ni^{+2}
(c) Co^{+3} (d) Fe^{+2}
96. Among the following outermost configurations of transition metals, which shows the highest oxidation state
(a) $3d^3 4s^2$ (b) $3d^5 4s^1$
(c) $3d^5 4s^2$ (d) $3d^6 4s^2$
97. Which of the following is not true for transition metals
(a) They are malleable and ductile
(b) They have high boiling and melting points
(c) They crystallize with body centred cubic and hexagonal close-packed structures only
(d) They show variable oxidation states although not always
98. The most malleable of all the metals is
(a) Silver (b) Sodium
(c) Gold (d) Platinum
99. Paramagnetism is exhibited by molecules
(a) Not attracted in a magnetic field
(b) Containing only paired electrons
(c) Carrying a positive charge
(d) Carrying unpaired electrons
100. The higher number of unpaired electrons are in
(a) Fe (b) Fe^+
(c) Fe^{+2} (d) Fe^{+3}

